

Supply Chain Dataset Analysis

Project Documentation

DEPI Data Analytics Program

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Project Overview

This project analyzes a 100-SKU supply chain dataset spanning cosmetics, haircare, and skincare products manufactured and shipped across five sourcing hubs in India. The analysis covers data cleaning, exploratory analysis, predictive modeling, and an interactive Power BI dashboard built on the cleaned dataset.

Project Duration: 4 Weeks

Tools Used:

- Python (pandas, NumPy, scikit-learn, Matplotlib, Seaborn)
- Google Colab
- Power BI Desktop

Dataset Snapshot

Property	Value
Rows (SKUs)	100
Columns (raw)	24
Product categories	Cosmetics, Haircare, Skincare
Sourcing locations	Mumbai, Delhi, Kolkata, Bangalore, Chennai
Suppliers	5 (Supplier 1 – Supplier 5)

Project Timeline

Week	Task	Deliverable
Week 1	Data Cleaning & Preprocessing	Cleaned CSV file, validation checks
Week 2	Analysis Phase	7 analytical questions answered with charts
Week 3	Forecasting	Predictive models (Random Forest, classification)
Week 4	Power BI Dashboard	Interactive dashboard, final documentation

Week 1: Data Cleaning & Preprocessing

1. Project Planning

1.1 Project Objectives

Objective	Description
Data Cleaning	Validate and preprocess the raw 100-row supply chain dataset
Feature Engineering	Derive Profit, Profit Margin %, Revenue per Unit, and a defect flag
Business Analysis	Answer 7 analytical questions covering revenue, suppliers, shipping, and quality
Forecasting	Predict revenue, shipment cost, and inspection outcome using scikit-learn
Interactive Dashboard	Build a Power BI dashboard for supply chain stakeholders

1.2 Tools Used

Tool	Purpose
Python (pandas, NumPy)	Data cleaning and feature engineering
Matplotlib / Seaborn	Visual analysis (Week 2)
scikit-learn	Forecasting models (Week 3)
Power BI Desktop	Interactive dashboard (Week 4)

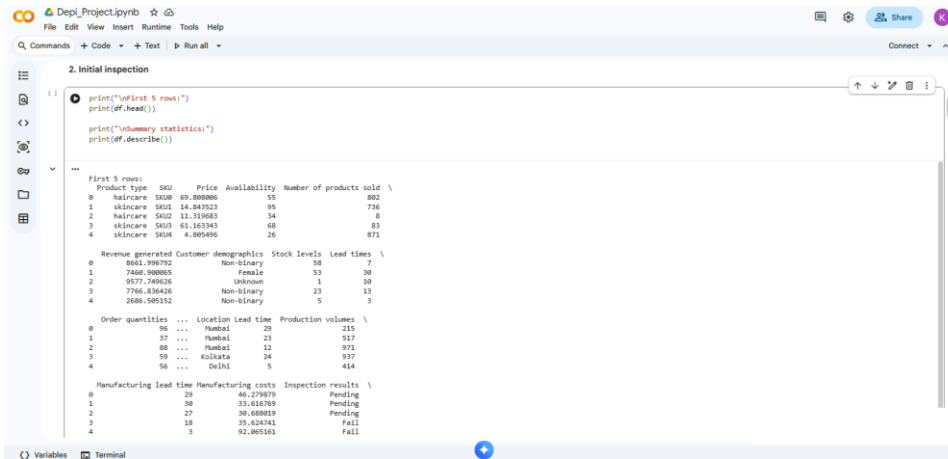
2. Data Cleaning Process (Python)

2.1 Steps Performed

Step	Description	Code/Method
1. Load Data	Read the raw CSV/Excel file	<code>pd.read_csv()</code> / <code>pd.read_excel()</code>
2. Initial Inspection	Check shape, first rows, and summary statistics	<code>df.head()</code> , <code>df.describe()</code>
3. Missing Values	Check for nulls in every column	<code>df.isnull().sum()</code>
4. Duplicate Rows	Check for duplicate rows and duplicate SKUs	<code>df.duplicated().sum()</code>
5. Invalid Values	Scan numeric columns for negative or out-of-range values	<code>(df[col] < 0).sum()</code>
6. Standardize Text	Strip whitespace and inspect unique categorical values	<code>df[col].str.strip()</code>
7. Outlier Scan	IQR-based outlier scan on key numeric fields	Q1/Q3 quantile, 1.5×IQR rule
8. Feature Engineering	Add Profit, Profit Margin %, Revenue per Unit, defect flag	Derived columns
9. Export Clean Data	Save the cleaned dataset	<code>df.to_csv()</code>

2.2 Python Code Snapshots

Snapshot 1: Initial Inspection



```

2. Initial Inspection

print("\nFirst 5 rows:")
print(df.head())

print("\nSummary statistics:")
print(df.describe())

'''
First 5 rows:
Product type  SKU  Price  Availability  Number of products sold \
0  skincare  SKU0  69.800000  55  802
1  skincare  SKU1  14.843923  95  736
2  skincare  SKU2  11.100603  24  8
3  skincare  SKU3  61.163343  68  83
4  skincare  SKU4  4.805496  26  871

Revenue generated Customer demographics Stock levels Lead times \
0  8861.590792  Non-binary  58  7
1  7468.900065  Female  53  30
2  9577.746426  Unknown  1  18
3  7786.836436  Non-binary  23  13
4  2686.505152  Non-binary  5  3

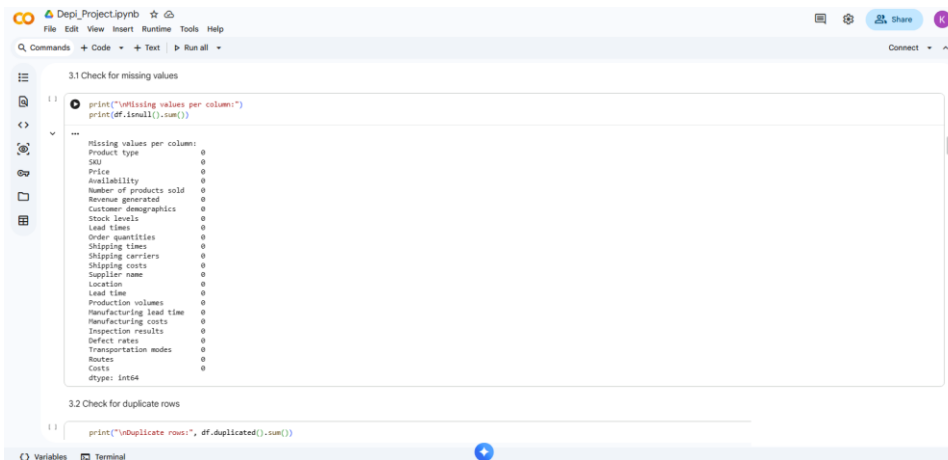
Order quantities ... Location Lead time Production volumes \
0  96 ... Mumbai  29  215
1  39 ... Mumbai  23  517
2  88 ... Mumbai  12  971
3  59 ... Kolkata  24  937
4  56 ... Delhi  5  414

Manufacturing lead time Manufacturing costs Inspection results \
0  29  40.278879  Pending
1  30  31.626708  Pending
2  27  30.688019  Pending
3  18  35.624742  Fail
4  3  92.805161  Fail
'''

```

Loading the dataset and reviewing the first five rows and summary statistics.

Snapshot 2: Missing Values Check



```

3.1 Check for missing values

print("\nMissing values per column:")
print(df.isnull().sum())

'''
Missing values per column:
Product type  0
SKU  0
Price  0
Availability  0
Number of products sold  0
Revenue generated  0
Customer demographics  0
Stock levels  0
Lead times  0
Order quantities  0
Shipping times  0
Shipping carriers  0
Shipping costs  0
Supplier name  0
Location  0
Lead time  0
Production volumes  0
Manufacturing lead time  0
Manufacturing costs  0
Inspection results  0
Defect rates  0
Transportation modes  0
Routes  0
Costs  0
dtype: int64
'''

3.2 Check for duplicate rows

print("\nDuplicate rows:", df.duplicated().sum())

```

Checking for missing values across all 24 raw columns — result: zero missing values.

Snapshot 3: Duplicates and Invalid Values

```

3.2 Check for duplicate rows
1) print("Duplicate rows:", df.duplicated().sum())
   print("Duplicate SKUs:", df['SKU'].duplicated().sum())

Duplicate rows: 0
Duplicate SKUs: 0

3.3 Check for invalid / out-of-range values
1) numeric_cols = df.select_dtypes(include=[np.number]).columns
   for col in numeric_cols:
       n_negative = (df[col] < 0).sum()
       if n_negative > 0:
           print(f"WARNING: {col} has {n_negative} negative values")
       print(f"No negative values found in numeric columns.")

No negative values found in numeric columns.

3.4 Standardize text/categorical columns (strip whitespace, fix casing)
1) categorical_cols = df.select_dtypes(include=['object', 'string']).columns
   for col in categorical_cols:
       df[col] = df[col].astype(str).strip()

   print("Unique values for key categorical columns:")
   for col in ['Product type', 'Customer demographics', 'Shipping carriers',
               'Supplier name', 'Location', 'Inspection results',
               'Transportation modes', 'Routes']:
       print(f"{col}: {df[col].unique().tolist()}")
  
```

Checking for duplicate rows/SKUs, scanning for negative values, and standardizing text columns.

Snapshot 4: Outlier Scan and Feature Engineering

```

3.5 Outlier check using IQR method
1) def iqr_outlier_report(series, name):
    q1, q3 = series.quantile([0.25, 0.75])
    iqr = q3 - q1
    lower, upper = q1 - 1.5 * iqr, q3 + 1.5 * iqr
    outliers = series[(series < lower) | (series > upper)]
    print(f"({name}): {len(outliers)} potential outliers "
          f"({range checked: {lower:.2f} to {upper:.2f}})")

    print(f"Outlier scan (IQR method):")
    for col in ['Price', 'Revenue generated', 'Shipping costs',
               'Manufacturing costs', 'Defect rates', 'Costs']:
        iqr_outlier_report(df[col], col)

Outlier scan (IQR method):
Price: 0 potential outliers (range checked: -66.00 to 363.00)
Revenue generated: 0 potential outliers (range checked: -3040.85 to 36415.67)
Shipping costs: 0 potential outliers (range checked: -2.55 to 13.49)
Manufacturing costs: 0 potential outliers (range checked: -45.47 to 137.00)
Defect rates: 0 potential outliers (range checked: -2.82 to 7.40)
Costs: 0 potential outliers (range checked: -347.67 to 3429.53)

4. Build a simple data model
Profit margin per unit sold = Revenue - (Manufacturing cost * units) - shipping/transport cost

1) df['Total manufacturing cost'] = df['Manufacturing costs'] * df['Production volumes']
   df['Profit'] = df['Revenue generated'] - df['Costs']
   df['Profit margin %'] = (df['Profit'] / df['Revenue generated']) * 100

Revenue per unit sold
  
```

IQR-based outlier scan on six numeric fields, followed by the start of feature engineering.

Snapshot 5: Engineered Columns and Export

The screenshot shows a Jupyter Notebook with the following code and output:

```
df['total manufacturing cost'] = df['manufacturing costs'] * df['production volumes']
df['profit'] = df['revenue generated'] - df['costs']
df['profit margin %'] = (df['profit'] / df['revenue generated']) * 100
```

Revenue per unit sold

```
df['revenue per unit'] = df['revenue generated'] / df['number of products sold']
```

Defect flag

```
df['has defect issue'] = df['defect rates'] > df['defect rates'].median()
print("New engineered columns added:")
print(df[['profit', 'profit margin %', 'revenue per unit', 'has defect issue']].head())
```

New engineered columns added:

	profit	profit margin %	revenue per unit	has defect issue
0	8474.244737	97.832462	16.800495	False
1	6957.834486	93.257307	10.137092	True
2	9425.820344	98.538229	1397.118793	True
3	7312.068266	96.719892	93.576342	True
4	1763.064328	69.426694	3.084392	True

5. Save cleaned dataset

```
df.to_csv('supply_chain_data_cleaned.csv', index=False)
df.to_excel('supply_chain_data_cleaned.xlsx', index=False)
print("Cleaned dataset saved as 'supply_chain_data_cleaned.csv' and '.xlsx'")
print("Final shape:", df.shape)
```

Cleaned dataset saved as 'supply_chain_data_cleaned.csv' and '.xlsx'

Adding Revenue per Unit and a defect flag, then exporting the cleaned dataset to CSV and Excel.

2.3 Key Cleaning Results

Check	Result
Missing values	0 across all 24 columns
Duplicate rows	0
Duplicate SKUs	0
Negative values	0 found in any numeric column
Outliers (IQR method)	0 flagged across 6 key numeric fields

The dataset arrived in an unusually clean state. No imputation, deduplication, or outlier removal was required — the only preprocessing steps were text standardization and the addition of engineered features described below.

2.4 Engineered Features

Feature	Formula	Purpose
Profit	Revenue generated – Costs	Net profit per SKU
Profit margin %	$(\text{Profit} \div \text{Revenue generated}) \times 100$	Profitability ratio
Revenue per unit	Revenue generated $\div$ Number of products sold	Per-unit pricing sanity check
Has defect issue	Defect rates > median(Defect rates)	Binary quality flag for segmentation

Week 2: Analysis Phase

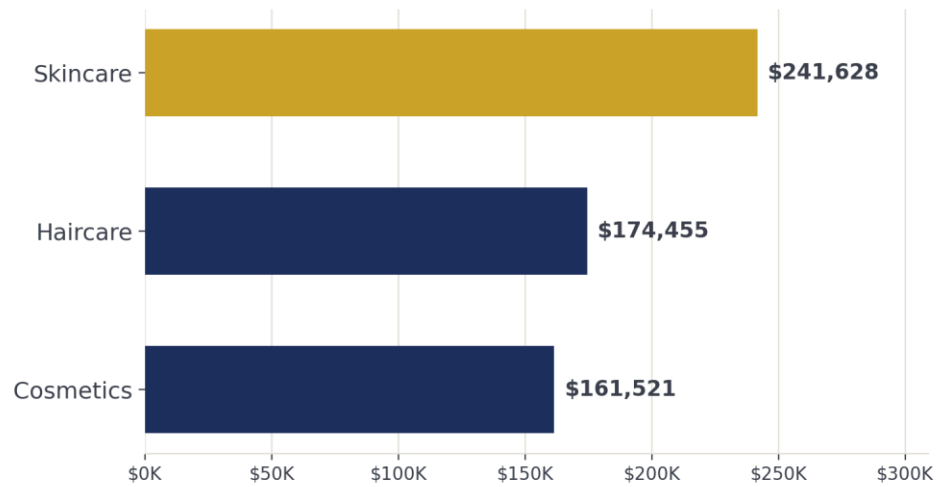
Seven analytical business questions were answered using the cleaned dataset, covering revenue, suppliers, shipping, location, pricing, and quality.

3.1 Query Categories

Category	Questions
Revenue & Products	Q1, Q5
Suppliers & Quality	Q2, Q7
Shipping & Logistics	Q3, Q4
Locations	Q6

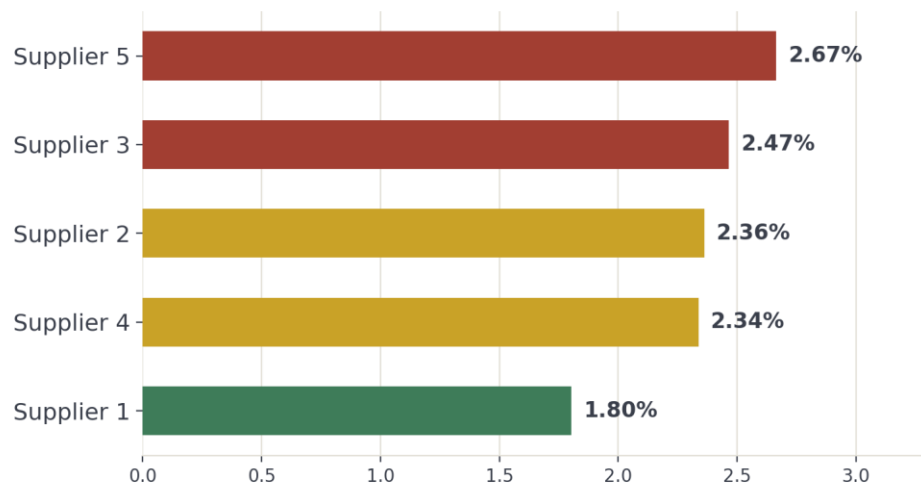
3.2 Questions, Results, and Insights

Q1: What is the impact of product type on revenue?



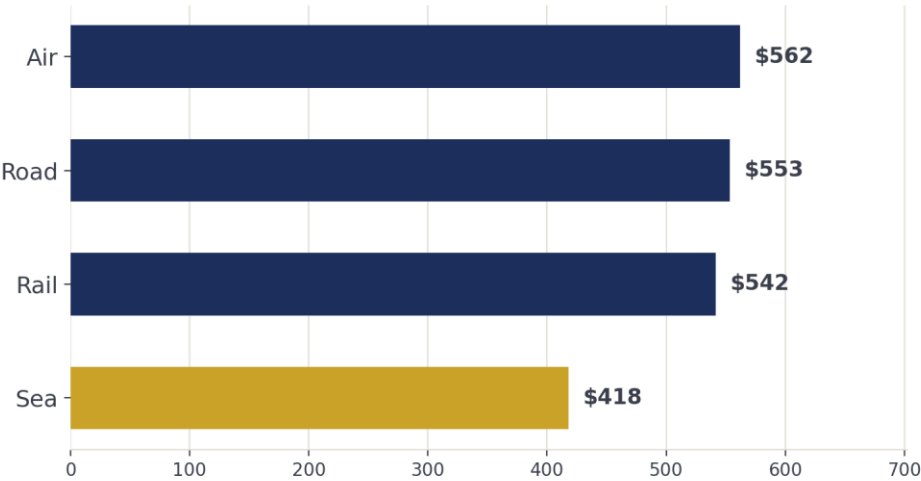
Total revenue by product category.

Skincare leads with \$241,628 in revenue (40 SKUs), followed by Haircare at \$174,455 (34 SKUs) and Cosmetics at \$161,521 (26 SKUs). Skincare also leads on unit volume with 20,731 units sold.

Q2: Which supplier contributes most to defect rates?*Average defect rate by supplier.*

Supplier 1 is the quality benchmark at 1.80% average defect rate. Supplier 5 runs the highest at 2.67% — 48% higher than Supplier 1.

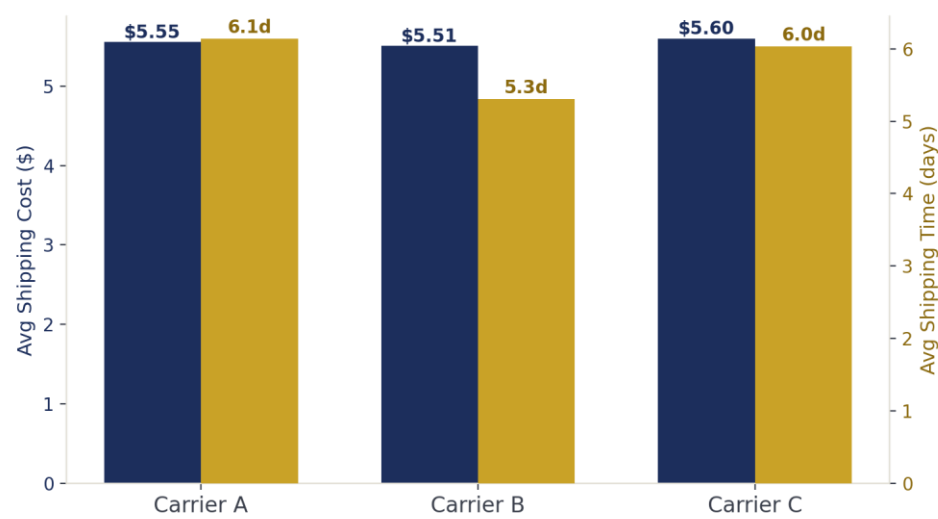
Q3: Which transportation mode is the most cost-efficient?



Average total cost by transportation mode.

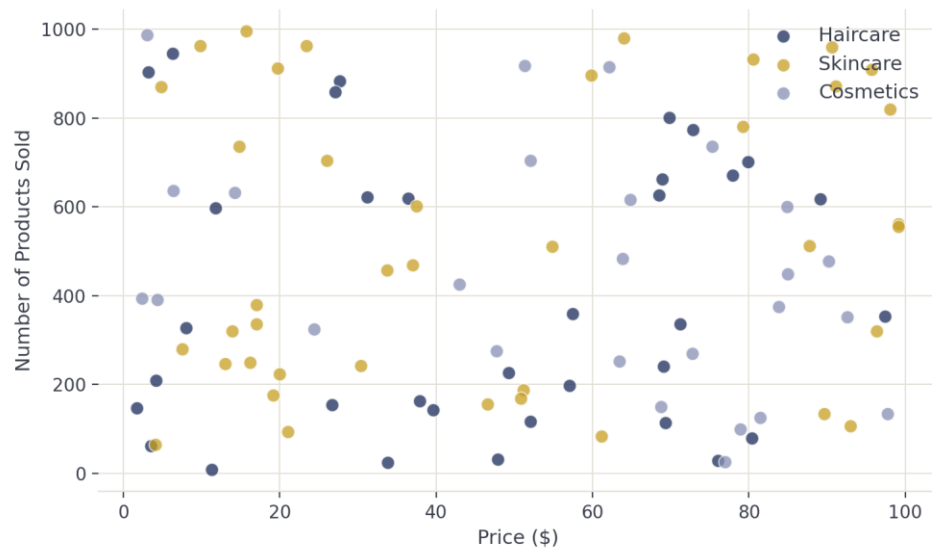
Sea freight averages \$417.82 per shipment — 25% cheaper than Air (\$561.71) — yet only 17 of 100 SKUs ship by sea, indicating an underused cost-saving opportunity.

Q4: How do shipping carriers compare in cost and time?



Average shipping cost and shipping time by carrier.

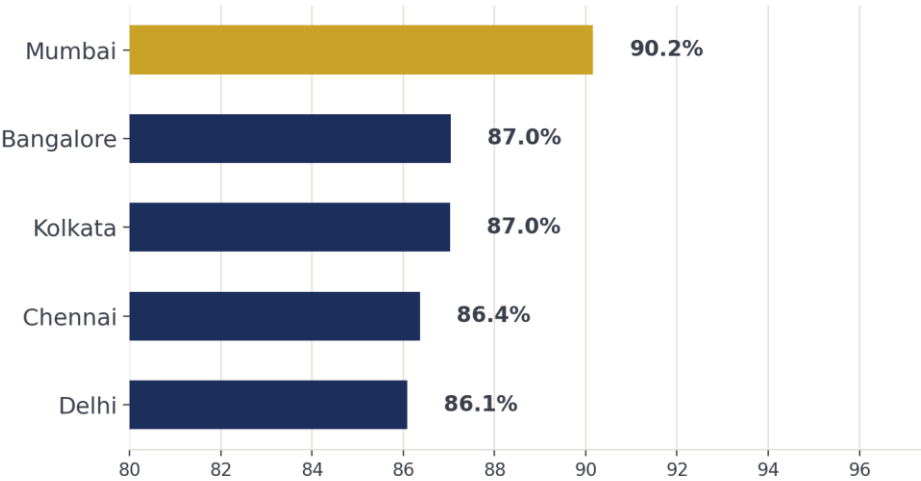
Carrier B is both the cheapest (\$5.51 average) and the fastest (5.3 days average), making it the strongest default choice among the three carriers.

Q5: Is there a relationship between price and units sold?

Price vs. number of products sold, colored by product type.

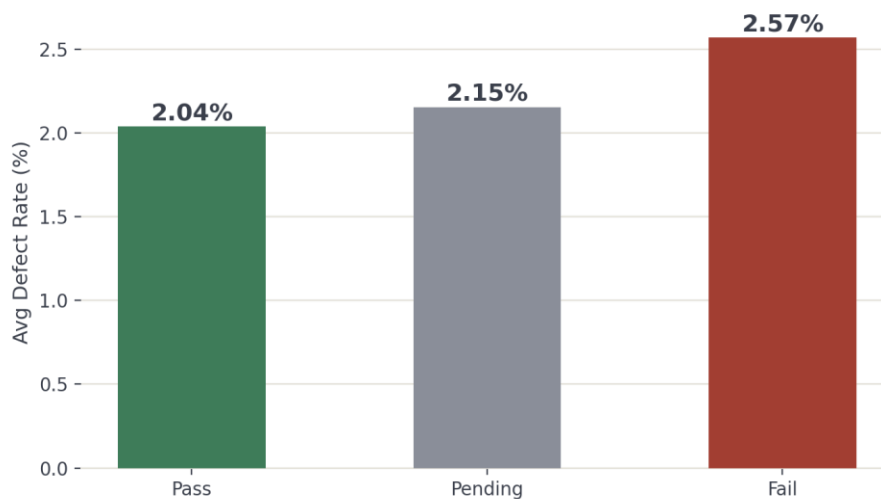
The correlation between price and units sold is just 0.006 — effectively no relationship. Pricing does not appear to drive sales volume in this catalog.

Q6: Which location generates the highest profit margin?



Average profit margin % by sourcing location.

Mumbai leads with a 90.2% average profit margin, outperforming every other hub by 3 or more margin points (Bangalore and Kolkata: 87.0%, Chennai: 86.4%, Delhi: 86.1%).

Q7: How does inspection result relate to defect rate?*Average defect rate by inspection outcome.*

Failed inspections average 2.57% defects versus 2.04% for passed inspections — a real but modest gap. The inspection outcome distribution itself (see Section 5) shows a larger story: only 23% of shipments pass on the first attempt.

3.3 Key Findings from Analysis

Finding	Value
Total Revenue	\$577,604.82
Total Profit	\$524,680.24
Average Profit Margin	87.44%
Top Category	Skincare (\$241,628)
Top Location	Mumbai (90.2% margin)
Cheapest Transport Mode	Sea (\$417.82 avg.)
Best Supplier (lowest defects)	Supplier 1 (1.80%)
QC Pass Rate	23.0%

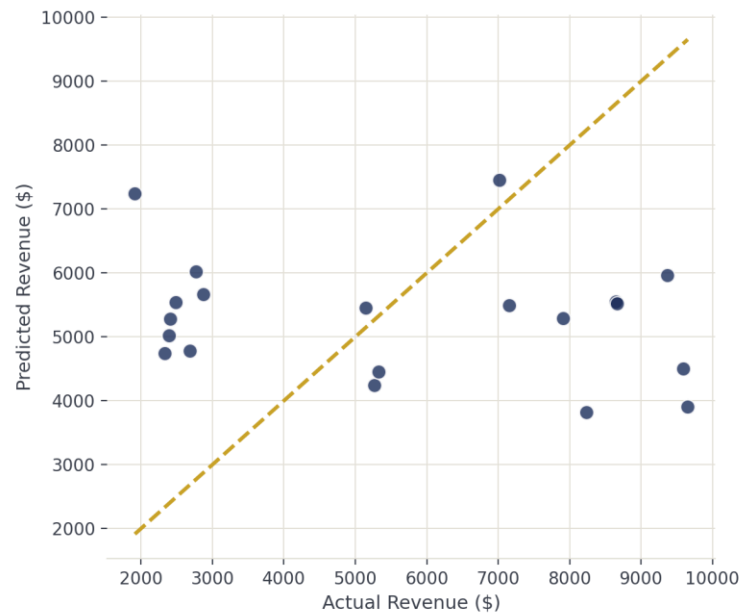
Week 3: Forecasting

This dataset has no date or time column, so classic time-series forecasting (predicting future months) is not possible. Instead, predictive models were built to forecast an outcome for a new product or shipment based on its characteristics — the standard adaptation for this type of cross-sectional dataset. This still satisfies the requirement of using scikit-learn for predictive modeling.

4.1 Methodology

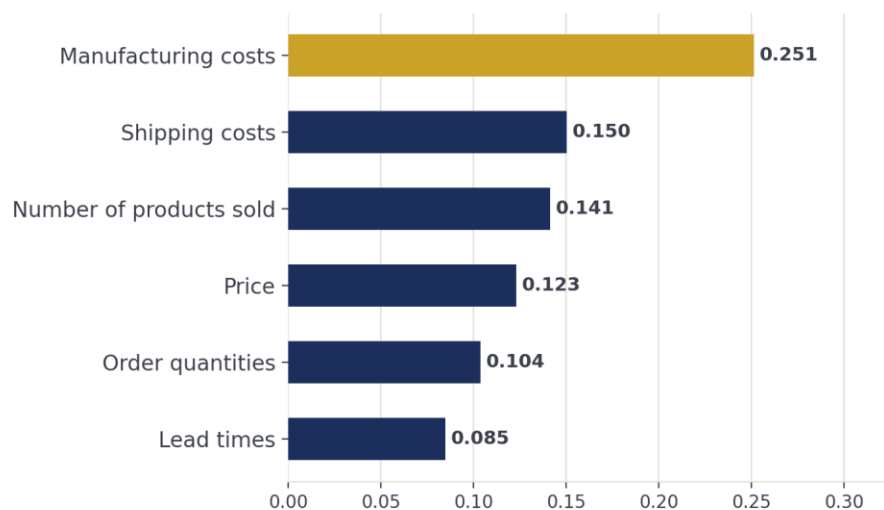
Component	Description
Models	Random Forest Regressor (revenue, cost); Random Forest Classifier (inspection outcome)
Data	100 SKUs, 80/20 train-test split
Target 1	Revenue generated
Target 2	Total shipment cost
Target 3	Inspection failure (binary classification)
Features	Price, units sold, order quantities, lead times, shipping/manufacturing costs, product type, customer demographics, supplier, transportation mode, route

4.2 Forecasting Q1: Predict Revenue Generated



Predicted vs. actual revenue. Points far from the diagonal line indicate poor fit.

Metric	Value
MAE (Mean Absolute Error)	\$2,813.60
R ² score	-0.262

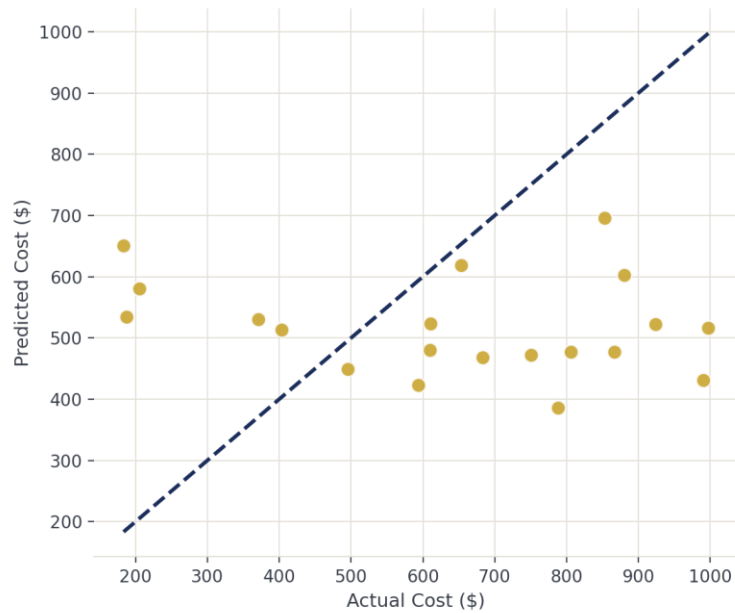


Top features driving the revenue forecast (Random Forest feature importance).

Manufacturing cost is the strongest individual driver of the revenue forecast, followed by shipping costs and number of products sold. However, the negative R² score indicates the model performs

worse than simply predicting the average — these importances should be read as directional signals, not as evidence of an accurate forecast.

4.3 Forecasting Q2: Predict Total Shipment Cost

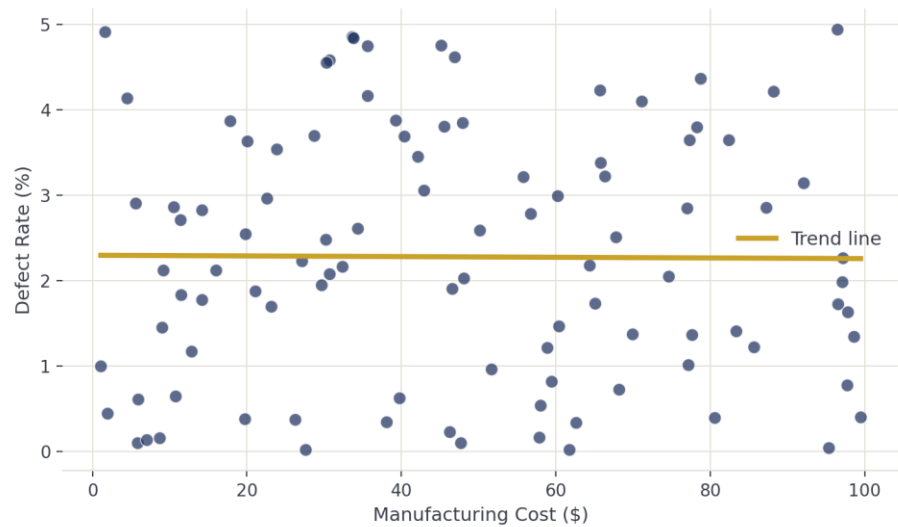


Predicted vs. actual shipment cost.

Metric	Value
MAE (Mean Absolute Error)	\$270.76
R ² score	-0.476

As with the revenue model, the cost model shows weak fit. This is consistent with the correlation analysis in Section 4.5 below.

4.4 Forecasting Q3: Predict Inspection Failure (Classification)



Manufacturing cost vs. defect rate, with a linear trend line. The near-flat slope confirms a weak relationship.

Metric	Value
Accuracy	55.0%
Precision (will-fail class)	0.40
Recall (will-fail class)	0.57
Test set size	20 shipments

The classification model performs modestly better than chance, suggesting defect rate and manufacturing cost carry a weak but real signal for quality risk. This is the most defensible forecasting result in this project — useful as an early-warning triage filter rather than a guarantee.

4.5 Interpretation Note: Why the Forecasts Are Weak

The R^2 scores for the revenue and cost models are low or negative. This is an expected and important finding, not a coding error. Two factors explain it:

- Sample size: with only 100 rows and an 80/20 split, the training set has just 80 observations — too few for a Random Forest to learn a robust pattern.
- Weak correlation: the strongest predictor of Revenue generated (excluding Profit-derived columns) correlates at only $|r| \approx 0.23$. This is a known characteristic of this public dataset, which was generated for exploratory analysis practice rather than as a realistic, strongly-correlated business dataset.

Presenting this limitation transparently is itself a deliverable of Week 3: it demonstrates an understanding of when a model is and is not appropriate for a given dataset.

Week 4: Power BI Dashboard

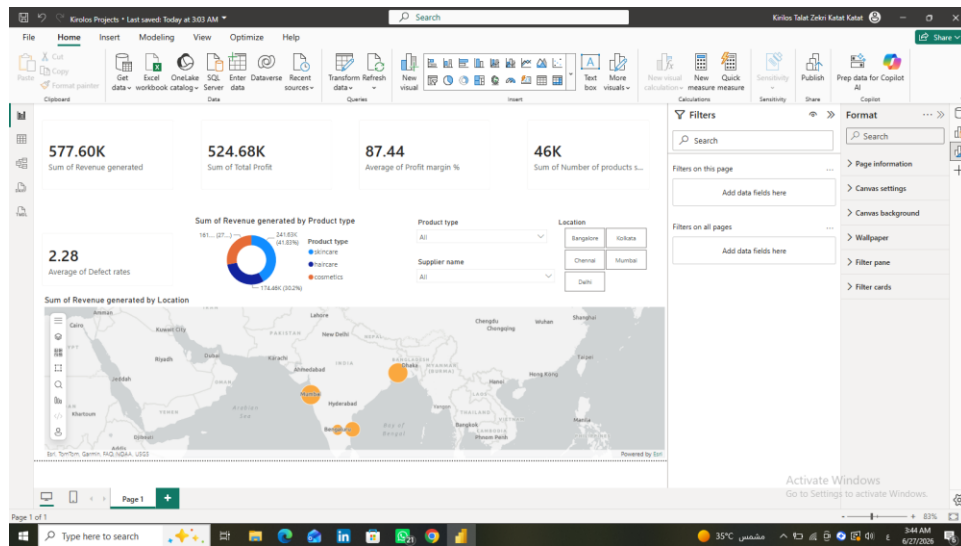
5.1 Dashboard Pages

Page	Content
Home	KPI cards, revenue by product type (donut), revenue by location (map), top-level slicers
Suppliers & Quality	Defect rate by supplier, inspection outcome breakdown, QC pass rate gauge

The dashboard follows a single consistent visual theme (navy and gold) across both pages, with synchronized slicers for Product type, Location, and Supplier name.

5.2 Dashboard Screenshots

Home Page



Home page: KPI cards, revenue donut by product type, slicers, and revenue-by-location map.

5.3 Key Measures (DAX)

QC Pass Flag (calculated column):

QC Pass Flag = 'supply_chain_data_cleaned'[Inspection results] = "Pass"

QC Pass Rate % (measure):

*QC Pass Rate % = AVERAGE('supply_chain_data_cleaned'[QC Pass Flag]) * 100*

5.4 Dashboard Features

Feature	Description
Slicers	Filter by Product type, Location, and Supplier name
Cards	Total Revenue, Total Profit, Average Profit Margin %, Units Sold, Average Defect Rate
Donut chart	Revenue share by product type
Map	Revenue by sourcing location (Esri map visual)

5.5 KPI Definitions

KPI	Formula	Description
Total Revenue	SUM(Revenue generated)	Sum of revenue across all SKUs
Total Profit	SUM(Profit)	Sum of profit across all SKUs
Avg Profit Margin %	AVERAGE(Profit margin %)	Average profitability ratio
Units Sold	SUM(Number of products sold)	Total units sold across all SKUs
Avg Defect Rate	AVERAGE(Defect rates)	Average product defect rate
QC Pass Rate %	AVERAGE(QC Pass Flag) × 100	Share of shipments passing inspection on first try

5.6 Assumptions & Business Rules

- All 100 SKUs were included in every visual; the dataset contained no missing or invalid records to exclude.
- Profit margin % and Defect rate are row-level percentages and were aggregated using AVERAGE, not SUM, to keep the resulting KPI meaningful.
- QC Pass Rate % is based strictly on the Inspection results field equaling “Pass” — “Pending” inspections are not counted as passes.
- Location values were assigned the “City” data category in Power BI to enable map plotting.

6. Key Findings Summary

6.1 Executive Summary

Metric	Value
Total Revenue	\$577,604.82
Total Profit	\$524,680.24
Average Profit Margin	87.44%
Total Units Sold	46,099
Average Defect Rate	2.28%
QC Pass Rate	23.0%
Average Lead Time	17.08 days
Average Shipping Cost	\$5.55 per unit
Total SKUs	100

6.2 Top Insights

Insight	Finding
Best Category	Skincare (\$241,628 revenue, 40 SKUs)
Best Location	Mumbai (90.2% average profit margin)
Best Supplier	Supplier 1 (1.80% average defect rate)
Worst Supplier	Supplier 5 (2.67% average defect rate, +48% vs. Supplier 1)
Cheapest Transport Mode	Sea (\$417.82 average, 25% cheaper than Air)
Best Carrier	Carrier B (cheapest and fastest)
Price–Sales Correlation	0.006 (effectively none)
Quality Bottleneck	Process, not product — 23% first-pass rate despite a 2.28% defect rate

7. Recommendations

Recommendation	Priority	Expected Impact
Shift more volume to Sea freight where lead time allows	High	Reduce logistics cost (Sea is 25% cheaper than Air)
Investigate and fix the inspection bottleneck (only 23% first-pass rate)	High	Faster throughput without compromising quality
Audit Supplier 5's process against Supplier 1's benchmark	Medium	Reduce defect rate by up to 48%
Prioritize Skincare SKUs sourced through Mumbai	Medium	Compound the two highest-performing segments
Treat forecasting outputs as directional only, not precise predictions	Low	Avoid over-reliance on a model with weak statistical fit

8. Project Limitations

- The dataset contains only 100 SKUs, limiting the statistical power of any predictive model.
- No date or time column exists, so true time-series forecasting was not possible — forecasts predict outcomes for new products/shipments based on attributes, not future time periods.
- Correlations between most numeric features and Revenue generated are weak ($|r| < 0.25$), which limits forecasting accuracy regardless of model choice.
- The dataset appears to be synthetically generated for exploratory analysis practice, so some relationships that would exist in real operational data (e.g., between price and demand) may not be present.
- “Pending” inspection results could not be resolved to Pass/Fail, so the true final pass rate may differ from the 23% first-pass figure reported here.

9. Final Project Files Structure

```
Supply_Chain_Project/ |— Data_Cleaning/ |   |— week1_data_cleaning.py |
|— supply_chain_data_cleaned.csv | |— Analysis/ |   |—
week2_analysis_questions.py | |— Forecasting/ |   |—
week3_forecasting_questions.py | |— PowerBI_Dashboard/ |   |—
Supply_Chain_Dashboard.pbix | |— Documentation/ |   |—
Supply_Chain_Documentation.docx    # This document
```

10. Conclusion

This project successfully:

1. Cleaned and validated a 100-SKU supply chain dataset, confirming zero missing values, duplicates, or outliers.
2. Engineered four new features (Profit, Profit Margin %, Revenue per Unit, defect flag) to support analysis.
3. Answered 7 analytical business questions covering revenue, suppliers, shipping, location, pricing, and quality.
4. Built and honestly evaluated three forecasting models using scikit-learn, including a transparent discussion of their limitations.
5. Built an interactive Power BI dashboard with KPI cards, a donut chart, a map, and synchronized slicers.

The analysis provides stakeholders with concrete, data-backed insights into:

- Revenue performance by product category and location
- Supplier quality and where defect rates concentrate
- Logistics cost efficiency across transportation modes and carriers
- The true nature of the quality bottleneck — an inspection process issue, not a manufacturing defect issue

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